

Nova Network and Environmental Qualification Report

Report scope

Requirements evaluated: 12

Domains: Communications, Environmental qualification

Methods: bench test, environmental exposure, inspection, log review

Result vocabulary: PASS, FAIL, NOT TESTED, INCONCLUSIVE

1. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-NET-001 CAN heartbeat	PASS - Executed for REQ-NVA-NET-001; the recorded result confirms: CAN FD heartbeat interval does not exceed 100 ms. FAIL - Executed for REQ-NVA-NET-001; the observed result does not satisfy: Heartbeat behavior is maintained for at least 30 minutes.	Bench / inspection record Reviewed by DV team
REQ-NVA-NET-002 Packet loss	NOT TESTED - No test or inspection was performed for REQ-NVA-NET-002: Heartbeat test duration is 30 minutes. NOT TESTED - No test or inspection was performed for REQ-NVA-NET-002: CAN bus load is 70 percent. NOT TESTED - No test or inspection was performed for REQ-NVA-NET-002: Lost heartbeat frame count is zero.	Bench / inspection record Reviewed by DV team
REQ-NVA-NET-003 Ethernet latency	INCONCLUSIVE - Work was attempted for REQ-NVA-NET-003, but incomplete scope or calibration prevents deciding: Diagnostic traffic is authenticated. INCONCLUSIVE - Work was attempted for REQ-NVA-NET-003, but incomplete scope or calibration prevents deciding: Transport uses 100BASE-T1. INCONCLUSIVE - Work was attempted for REQ-NVA-NET-003, but incomplete scope or calibration prevents deciding: Network utilization is 95 percent. INCONCLUSIVE - Work was attempted for REQ-NVA-NET-003, but incomplete scope or calibration prevents deciding: Diagnostic round-trip latency remains below 12 ms.	Bench / inspection record Reviewed by DV team
REQ-NVA-NET-004 Diagnostic session timeout	PASS - Executed for REQ-NVA-NET-004; the recorded result confirms: Extended diagnostic session is inactive. PASS - Executed for REQ-NVA-NET-004; the recorded result confirms: Session returns to default after 5 seconds. PASS - Executed for REQ-NVA-NET-004; the recorded result confirms: Temporary unlock state is cleared. PASS - Executed for REQ-NVA-NET-004; the recorded result confirms: Timeout event is recorded.	Bench / inspection record Reviewed by DV team

2. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-NET-005 Bus-off recovery	PASS - Executed for REQ-NVA-NET-005; the recorded result confirms: Bus-off is detected. IN PROGRESS - Partial execution exists for REQ-NVA-NET-005, but verification is not complete: Transmission is inhibited for at least 1.0 second. NOT TESTED - No test or inspection was performed for REQ-NVA-NET-005: Node recovers without an ignition cycle. NOT TESTED - No test or inspection was performed for REQ-NVA-NET-005: Persistent bus-off counter is incremented.	Bench / inspection record Reviewed by DV team
REQ-NVA-ENV-001 Ingress protection	NOT TESTED - No test or inspection was performed for REQ-NVA-ENV-001: Connector J17 is immersed to a depth of 1 m. NOT TESTED - No test or inspection was performed for REQ-NVA-ENV-001: Immersion duration is 30 minutes. NOT TESTED - No test or inspection was performed for REQ-NVA-ENV-001: Connector J17 retains IP67 protection.	Bench / inspection record Reviewed by DV team
REQ-NVA-ENV-002 Vibration endurance	INCONCLUSIVE - Work was attempted for REQ-NVA-ENV-002, but incomplete scope or calibration prevents deciding: Random vibration level is 30 g RMS. INCONCLUSIVE - Work was attempted for REQ-NVA-ENV-002, but incomplete scope or calibration prevents deciding: Exposure lasts 8 hours per axis. INCONCLUSIVE - Work was attempted for REQ-NVA-ENV-002, but incomplete scope or calibration prevents deciding: Electrical continuity is not lost.	Bench / inspection record Reviewed by DV team
REQ-NVA-ENV-003 Mechanical shock	PASS - Executed for REQ-NVA-ENV-003; the recorded result confirms: Three 50 g half-sine shocks are applied on each axis. PASS - Executed for REQ-NVA-ENV-003; the recorded result confirms: No fastener loosens. PASS - Executed for REQ-NVA-ENV-003; the recorded result confirms: No enclosure crack is longer than 2 mm.	Bench / inspection record Reviewed by DV team

3. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-ENV-004 Salt-mist resistance	PASS - Executed for REQ-NVA-ENV-004; the recorded result confirms: Salt-mist exposure lasts 96 hours. IN PROGRESS - Partial execution exists for REQ-NVA-ENV-004, but verification is not complete: Contact resistance remains below 15 mOhm. NOT TESTED - No test or inspection was performed for REQ-NVA-ENV-004: Sealing surfaces show no red corrosion. NOT TESTED - No test or inspection was performed for REQ-NVA-ENV-004: Labels remain legible.	Bench / inspection record Reviewed by DV team
REQ-NVA-ENV-005 Mounting integrity	PASS - Executed for REQ-NVA-ENV-005; the recorded result confirms: All four mounting bolts are included. PASS - Executed for REQ-NVA-ENV-005; the recorded result confirms: Initial mounting-bolt torque is 22 Nm. PASS - Executed for REQ-NVA-ENV-005; the recorded result confirms: Every mounting bolt has torque witness paint. FAIL - Executed for REQ-NVA-ENV-005; the observed result does not satisfy: Post-vibration residual torque is at least 18 Nm.	Bench / inspection record Reviewed by DV team

4. Image-based inspection result

Visual Inspection Record

REQ-NVA-NET-006 | Gateway routing

PASS

the recorded result confirms: A message with identifier 0x18FF50E5 enters port A.

PASS

the recorded result confirms: Message leaves through port C.

PASS

the recorded result confirms: Routing latency does not exceed 4 ms.

PASS

the recorded result confirms: Message payload is retained.

FAILED

the observed result does not satisfy: Alive counter is updated.

REQ-NVA-ENV-006 | Drain orientation

UNTESTED

NOT TESTED - No test or inspection was performed for REQ-NVA-ENV-006: Enclosure is mounted in vehicle orientation.

UNTESTED

NOT TESTED - No test or inspection was performed for REQ-NVA-ENV-006: Drain slot faces downward.

UNTESTED

NOT TESTED - No test or inspection was performed for REQ-NVA-ENV-006: Drain slot remains outside the gasket boundary.

UNTESTED

NOT TESTED - No test or inspection was performed for REQ-NVA-ENV-006: Unobstructed drain opening is at least 18 mm2.

Image record NOVA-DV-VIS | Reviewer: QA-17 | Controlled copy

Figure 4-1. Recorded inspections for REQ-NVA-NET-006, REQ-NVA-ENV-006; interpretation requires the raster content.